Suggested Teaching Guidelines for

Fundamentals of Artificial Intelligence & Mathematics for Artificial Intelligence

PGCP-AI February 2026

Duration: 60 Theory Hours

Objective: To introduce the student to fundamentals of Artificial Intelligence.

Prerequisites: Knowledge of programming language, Computing Fundamentals, and some basic statistical knowledge.

Evaluation method: Theory exam– 80% weightage
Internal exam– 20% weightage

List of Books / Other training material

Courseware:
1. Artificial Intelligence: A Modern Approach, 4e, Peter Norvig, Stuart J. Russell, PearsonEducation, 4th Edition.

Reference Book:
1. Artificial Intelligence by Example, Denis Rothman
2. Artificial Intelligence by Saroj Kaushik

Note:
- ***Each session mentioned is of 2 hours' theory. Faculty can give case studies to students***
- ***Faculties are advised to relate the topics with real world applications.***

Fundamentals of AI

Session 1:

Theory:
- Definition and scope of AI
- AI evolution and historical milestones
- Alan Turing's work
- Turing Machine
- Turing Test

Session 2:

Theory:
- Ethics of AI
- Being human in the age of AI
- Responsible AI
- Social, legal, and ethical implications

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Session 3 & 4:
Theory:

- Structure of AI systems
- Real-world implications of AI
- AI revolution
- Current trends in Artificial Intelligence

Session 5 & 6:
Theory:

- Artificial life
- Learning through interaction
- Emergent behavior
- Rules-based systems
- Expert systems

Session 7 & 8:
Theory:

- Supervised learning
- Unsupervised learning
- Learning models and examples

Session 9:
Theory:

- Knowledge representation techniques
- Problem-solving approaches
- State-space representation

Session 10 & 11:
Theory:

- Types of search problems
- Classical search methodologies
- Beyond classical search
- Parallel search
- Search engines
- Adversarial search

Session 12 & 13:
Theory:

- Intelligent agents
- Uninformed search
- Constraint satisfaction search
- Combinatorial optimization problems

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Session 14:
Theory:

- Knowledge representation and automation
- Propositional logic
- Predicate logic
- Logic concepts
- Logic programming

Session 15:
Theory:

- Inference and resolution for problem solving
- Advanced problem-solving paradigm: Planning
- Introduction to Big Data
- Structured and unstructured data
- Relevance of Big Data in AI
- Data analysis and data analytics
- Applications of Big Data

Mathematics for AI

Linear Algebra

Session 1

- Vectors, Definition,
- Scalars, Addition, Scalar Multiplication
- Inner Product (Dot Product), Vector Projection
- Cosine Similarity, Orthogonal Vectors

Session 2 & 3

- Normal And Orthonormal Vectors
- Vector Norm, Vector Space
- Linear Combination, Linear Span
- Linear Independence, Basis Vectors

Assignments: Consider the vectors {[3, 0, 4], [−1, 0, 7], [2, 9, 11]} .Check that the vectors are linearly independent or not?

Session 4

- Linear Independence
- Basis and Rank
- Linear Mappings
- Affine Spaces

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Session 5

- Matrices Definition, Addition, Transpose
- Scalar Multiplication, Matrix Multiplication, Matrix Multiplication Properties
- Hadamard Product, Functions, Linear Transformation, Determinant, Identity Matrix,
- Invertible

Assignments:

$X = [0\ 1\ 3]^T$ and $Y = [2\ \ 4\ 0]^T$

- Find V = Subspace of X
- Find W = Subspace of Y
- Describe V intersection W ?

Session 6

- Matrix and Inverse, Rank, Trace
- Popular Type of Matrices- Symmetric, Diagonal, Orthogonal, Orthonormal
- Positive Definite Matrix
- Matrix Phylogeny
- Matrix Approximation

Session 7 & 8

- Eigen Values & Eigenvectors, Concept, Intuition, Significance
- How To Find Principle Component Analysis
- Concept, Properties, Applications
- Singular Value Decomposition

Assignments:

Find the Eigen values and eigenvectors for the matrix

$$\begin{bmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{bmatrix}$$

Calculus

Session 9 & 10

- Functions, Scalar Derivative, Definition, Intuition
- Common Rules Of Differentiation, Chain Rule
- Partial Derivatives, Gradient
- Concept, Intuition, Properties
- Directional Derivative

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Assignments: Let $f(x,y,z)=xyex2+z2-5$f(x,y,z)=xyex2+z2−5. Calculate the gradient of ff at the point (1,3,−2)(1,3,−2) and calculate the directional derivative $D_u f$Duf at the point (1,3,−2)(1,3,−2) in the direction of the vector v=(3,−1,4)v=(3,−1,4).

Session 11

- Gradients of Vector Valued Functions
- Gradient of Matrices
- Useful Identities for Computing Gradients
- Back propagation and Automatic Differentiation
- Linearization and Multivariate Taylor Series

Session 12 & 13

- Optimization Using Gradient Descent
- Constrained Optimization and Lagrange Multipliers
- Convex Optimization

Session 14 & 15

- Vector And Matrix Calculus
- How To Find Derivative Of Scalar-Valued
- Vector-Valued Function With Respect To Scalar, Vector , Four Combinations- Jacobian
- Gradient Algorithms, Local/Global Maxima and Minima
- Saddle Point, Convex Functions
- Gradient Descent Algorithms- Batch, Mini-Batch, Stochastic
- Performance Comparison